

LogiFlow QA Bug Report

LogiFlow — Comprehensive QA Bug Report

Tested On: <https://logi-flow-solution-challenge-2026.vercel.app/> **Date:** June 12, 2026 **Tester:** Automated QA (Antigravity Agent) **Scope:** Road Pipeline, Air Pipeline, Smart Shipment Planner **Viewports:** Desktop (1280×800), Mobile (375×812)

Executive Summary

Severity	Count
 Critical	3
 High	7
 Medium	12
 Low	8
Total	30

CRITICAL BUGS

BUG-001: Air Pipeline — No Same-City Validation (Backend)

CAUTION The Air pipeline backend (`air_routes.py`) has **zero validation** for same-city routes. Unlike the Road pipeline, which explicitly checks `source == destination` (line 52), the Air endpoint blindly forwards any pair to the pipeline.

- **Severity:** 🚨 Critical
 - **Pipeline:** Air
 - **Layer:** Backend (FastAPI)
 - **Steps to Reproduce:**
 1. Navigate to `/air`
 2. Enter "Mumbai" as both origin and destination
 3. Click "Optimize air route"
 - **Expected:** Error message: "Origin and destination cannot be the same city"
 - **Actual:** Request is sent to the backend, which processes it and may return nonsensical routes or waste compute.
 - **Root Cause:** [air_routes.py](#) — the `optimize_air` function has no `source == destination` guard, unlike [road_routes.py L52-60](#).
 - **Frontend Also Missing:** [AirInputForm.tsx](#) — `onSubmit` calls `runAirOptimize()` without any same-city check. Compare with [roadInputForm.tsx L401-417](#) which has comprehensive validation.
-

BUG-002: Air Pipeline — No Weight Validation (Frontend & Backend)

CAUTION The Air form and backend accept any weight value including 0, negative, and absurdly large numbers.

- **Severity:** 🚨 Critical
 - **Pipeline:** Air
 - **Layer:** Both Frontend and Backend
 - **Steps to Reproduce:**
 1. Navigate to `/air`
 2. Set weight to `0` or `-50`
 3. Click "Optimize air route"
 - **Expected:** Validation error for invalid weight
 - **Actual:** Request is sent to backend with invalid weight
 - **Root Cause:**
 - **Frontend:** [AirInputForm.tsx L98-104](#) — `onSubmit` only checks `source.trim()` and `destination.trim()`, never validates `cargoWeight`. The Road form checks `cargoWeight <= 0` at [roadInputForm.tsx L411-414](#).
 - **Backend:** [AirCargoPayload](#) has `cargo_weight_kg: float = 100` but no `field_validator` for range constraints.
-

BUG-003: Air Pipeline — Accepts Gibberish Cities and Returns Misleading Results

CAUTION Entering nonsensical city names (e.g., "xyz" and "abc") causes the autocomplete to show matching entries (ABC airport code), and submitting returns a "No air routes found" message — but the **dropdown shows duplicate entries** and the previous results remain visible below the error message, creating confusion.

- **Severity:**  Critical
 - **Pipeline:** Air
 - **Layer:** Frontend (Autocomplete + Results Persistence)
 - **Steps to Reproduce:**
 1. Navigate to `/air`
 2. Run a valid search (e.g., Mumbai → Delhi) to get results
 3. Change origin to "xyz" and destination to "abc"
 4. The autocomplete dropdown shows 4 identical "abc" / "ABC" entries
 5. Submit — "No air routes found" message appears
 6. Previous results (from Mumbai → Delhi) remain visible below
 - **Expected:** (a) Autocomplete should deduplicate entries. (b) Previous results should be cleared when new search begins. (c) Better validation before allowing gibberish submissions.
 - **Impact:** Users see stale results from a previous search mixed with an error banner, creating a deeply confusing UX.
-

HIGH SEVERITY BUGS

BUG-004: Road Pipeline — Map Zooms to World View Instead of Route

- **Severity:**  High
 - **Pipeline:** Road
 - **Layer:** Frontend (Map Component)
 - **Steps to Reproduce:**
 1. Navigate to `/road`
 2. Search "Mumbai" → "Delhi"
 3. Wait for results, then expand Cost Breakdown section
 4. Observe the map zooms out to show entire world (Australia, Philippines visible)
 - **Expected:** Map should stay focused on the Mumbai-Delhi corridor (India)
 - **Actual:** Map zooms out to world view after interacting with cost breakdown panel
 - **Impact:** Users lose visual context of the route when exploring cost details
-

BUG-005: Road Pipeline — No Auto-Scroll to Results

- **Severity:**  High
 - **Pipeline:** Road, Air (both affected)
 - **Layer:** Frontend (UX)
 - **Steps to Reproduce:**
 1. Navigate to `/road` (desktop viewport)
 2. Enter valid route and click "Find optimal routes"
 3. Wait for results to load
 - **Expected:** Page should auto-scroll to the results section
 - **Actual:** Results render below the fold. User must manually scroll down to see them.
 - **Impact:** On mobile especially, users may think the search failed because they see no immediate visual feedback when results load below the visible area.
-

BUG-006: Air Pipeline — Shimla-Puducherry Returns Routes (Should Return “No Routes”)

- **Severity:** 🟡 High
 - **Pipeline:** Air
 - **Layer:** Backend (Route Graph)
 - **Steps to Reproduce:**
 1. Navigate to `/air`
 2. Enter “Shimla” → “Puducherry”
 3. Submit — observe that 4 routes are returned with 94% confidence
 - **Expected:** These small cities have no direct airports. Should return “No air routes found” or “Limited airport access”.
 - **Actual:** Returns 4 ranked routes (all showing IndiGo, Direct flights) with suspiciously identical confidence scores (94%). This suggests the backend is **fabricating routes** or mapping to nearest airports without disclosure.
 - **Impact:** Completely misleading to logistics operators who may plan shipments based on non-existent air routes.
-

BUG-007: Air Pipeline — All Route Cards Show Identical Data (IndiGo, 94%, Direct)

- **Severity:** 🟡 High
 - **Pipeline:** Air
 - **Layer:** Backend + Frontend
 - **Steps to Reproduce:**
 1. Navigate to `/air`
 2. Search “Mumbai” → “Delhi”
 3. Observe route cards: all 4 show “IndiGo”, “Direct”, “94%”
 - **Expected:** Different route options should show different airlines, connection types, and confidence scores
 - **Actual:** All 4 route cards appear identical — same airline, same flight type, same confidence score. The only visible difference is the ranking number (#1, #2, etc.).
 - **Impact:** Users cannot make informed decisions between supposedly “different” route options. This defeats the purpose of multi-route ranking.
-

BUG-008: Notification Bell — No Visible Response on Click

- **Severity:** 🟡 High
 - **Pipeline:** Cross-cutting (NavBar)
 - **Layer:** Frontend (UX)
 - **Steps to Reproduce:**
 1. On any page, click the bell icon (🔔) in the top-right navbar
 - **Expected:** Either a notification dropdown, a "No notifications" message, or a redirect to a notifications page
 - **Actual:** Nothing happens. No visual feedback, no dropdown, no navigation.
 - **Impact:** The bell icon is a false affordance. Users will click it expecting notifications functionality and be confused by the lack of response.
-

BUG-009: Road Pipeline — "AI Explain" Button Truncation in Detail Panel

- **Severity:** 🟡 High
 - **Pipeline:** Road
 - **Layer:** Frontend (UI)
 - **Steps to Reproduce:**
 1. Navigate to `/road`, run a valid search
 2. In the route detail panel, click the "AI Explain" button
 3. Wait for the AI explanation to load
 - **Expected:** Full AI-generated explanation displayed
 - **Actual:** The explanation text is truncated or overlaps with surrounding elements. The container does not expand to accommodate the full text.
 - **Impact:** Users lose valuable AI insight that could help decision-making.
-

BUG-010: Autocomplete Dropdown Duplicates Across Pipelines

- **Severity:** 🟡 High
 - **Pipeline:** Air (confirmed), likely also Road
 - **Layer:** Frontend (Autocomplete)
 - **Steps to Reproduce:**
 1. Navigate to `/air`
 2. Type "Puducherry" in destination field
 3. Observe dropdown shows 3 identical "Puducherry" entries, all with code "PUDUC"
 - **Expected:** Single "Puducherry" entry or differentiated entries (e.g., different airports/stations)
 - **Actual:** Three identical entries with no distinguishing information
 - **Root Cause:** The `useCitySearch` hook or backend location API returns duplicates. The frontend does not deduplicate.
-

🟡 MEDIUM SEVERITY BUGS

BUG-011: Mobile — Navbar Hamburger Menu Exists but Smart Shipment Planner Button Competes

- **Severity:** 🟡 Medium
 - **Viewport:** Mobile (375×812)
 - **Steps to Reproduce:**
 1. Load any page on mobile viewport
 2. Observe navbar: hamburger (≡) on left, "+ Planner" button on right
 - **Expected:** Clean mobile navbar with all actions behind hamburger OR a minimal CTA
 - **Actual:** Both hamburger and "+ Planner" button visible. The Planner button text may truncate on smaller screens (320px width).
 - **Impact:** Visually cluttered navbar on small screens
-

BUG-012: Mobile — Form Fields Lack Minimum Touch Targets

- **Severity:** 🟡 Medium
 - **Viewport:** Mobile (375×812)
 - **Pipeline:** Road, Air
 - **Steps to Reproduce:**
 1. Load `/road` or `/air` on mobile
 2. Inspect ChoicePills (Cargo Type, Priority)
 3. Measure touch targets
 - **Expected:** All interactive elements $\geq 44 \times 44$ px (Apple/Google accessibility guidelines)
 - **Actual:** ChoicePill buttons are ~35px tall, making them difficult to tap accurately
 - **Impact:** Accessibility violation, poor usability on touch devices
-

BUG-013: Road Pipeline — Cargo Weight Allows 0 via Number Input Spin-Down

- **Severity:** 🟡 Medium
 - **Pipeline:** Road
 - **Layer:** Frontend
 - **Steps to Reproduce:**
 1. Navigate to `/road`
 2. Use the number input spinner to decrease weight below 1
 3. The `min={1}` HTML attribute prevents typing "0" but the validation at [roadInputForm.tsx L411](#) only checks `cargoWeight <= 0`, allowing 0 through the number input on some browsers
 - **Expected:** Weight should be validated to be > 0 on blur and on submit
 - **Actual:** In some browsers, `min={1}` is advisory not enforced, and typing "0" triggers the error but using the spinner may not.
-

BUG-014: Road Pipeline — “Start Driving” Opens Google Maps but Route May Differ

- **Severity:** 🟡 Medium
 - **Pipeline:** Road
 - **Layer:** Frontend (UX/Expectation Management)
 - **Steps to Reproduce:**
 1. Get road results, click “Start Driving”
 2. Google Maps opens with origin/destination
 - **Expected:** Disclaimer that Google Maps may choose different roads
 - **Actual:** Disclaimer exists (“LogiFlow sets the stop sequence · Google Maps chooses roads between stops”) but only for multi-stop routes. For simple A→B routes, there’s no warning.
 - **Impact:** Users may trust that the Google Maps route is identical to LogiFlow’s optimized route, leading to different cost/time than expected.
-

BUG-015: Air Pipeline — Window Slider Label Not Clearly Explained

- **Severity:** 🟡 Medium
 - **Pipeline:** Air
 - **Layer:** Frontend (UX)
 - **Steps to Reproduce:**
 1. Navigate to `/air`
 2. Look at the “Window · 48h” label and slider
 - **Expected:** Clear explanation of what “Window” means (delivery window? departure flexibility?)
 - **Actual:** Label says `Window · 48h` with no tooltip or helper text. The underlying code maps it to `deadlineHours` which represents a delivery deadline, but the label “Window” doesn’t convey this.
 - **Impact:** Users won’t understand what they’re adjusting
-

BUG-016: Road Pipeline — “More Options” Toggle Not Obvious

- **Severity:** 🟡 Medium
 - **Pipeline:** Road
 - **Layer:** Frontend (UX)
 - **Steps to Reproduce:**
 1. Navigate to `/road`
 2. Look for advanced options (Budget Cap, Deadline, Vehicle Type, Fuel Price)
 3. They are hidden behind “More options” text at the top-right of the form card
 - **Expected:** Advanced toggle should be more discoverable
 - **Actual:** Small “More options” text is easy to miss. Many users will never discover Budget Cap, Vehicle Type, and Fuel Price settings.
-

BUG-017: Road Pipeline — Cost Breakdown Total May Not Match Route Cost

- **Severity:** 🟡 Medium
 - **Pipeline:** Road
 - **Layer:** Frontend (Calculation)
 - **Root Cause:** [RouteResults.tsx L224-264](#) — `buildCostBreakdown()` uses **synthetic estimates** when backend doesn't provide all fields. The `total` is recomputed as `fuel + driver + toll + handling + stops + gst + documentation`, which may differ from `route.cost`.
 - **Example:** Route cost = ₹3,528 but breakdown total = ₹3,528 only when all synthetic multipliers align. For routes where backend provides partial data, the breakdown total can diverge.
 - **Impact:** Users comparing the headline cost with the breakdown total may notice discrepancies, eroding trust.
-

BUG-018: Login Page — No Loading State for Google OAuth

- **Severity:** 🟡 Medium
- **Pipeline:** Smart Shipment Planner
- **Layer:** Frontend (UX)
- **Steps to Reproduce:**
 1. Navigate to `/login`
 2. Click “Continue with Google”
- **Expected:** Button should show a loading spinner while OAuth redirect processes
- **Actual:** Button clicks but no visual feedback during the redirect. On slow connections, users may click multiple times.

BUG-019: Dashboard/Reports — Redirect Behavior for Unauthenticated Users Unclear

- **Severity:** 🟡 Medium
 - **Pipeline:** Smart Shipment Planner
 - **Layer:** Frontend (Auth Guard)
 - **Steps to Reproduce:**
 1. Navigate directly to `/dashboard` without being logged in
 2. Navigate directly to `/reports` without being logged in
 - **Expected:** Redirect to `/login` with a return URL parameter
 - **Actual:** Testing showed the pages redirect, but there's no toast message explaining "You must log in to access this page."
 - **Impact:** Users arriving via direct links won't understand why they were redirected.
-

BUG-020: Road Pipeline — Simulation Mode Always Visible (Should Be Behind "More Options")

- **Severity:** 🟡 Medium
 - **Pipeline:** Road
 - **Layer:** Frontend (UX)
 - **Steps to Reproduce:**
 1. Navigate to `/road`
 2. The Simulation Mode section with checkbox, presets, and sliders is always visible at the bottom of the form
 - **Expected:** Simulation Mode is a power-user feature and should be hidden behind "More Options" toggle
 - **Actual:** Always visible, making the form long and potentially overwhelming for first-time users
 - **Root Cause:** [roadInputForm.tsx L772](#) — Simulation Mode is rendered outside the `showAdvanced` conditional block.
-

BUG-021: Air Pipeline — “Understand & fill form” / “Understand & run optimize” Buttons Not Differentiated Enough

- **Severity:**  Medium
 - **Pipeline:** Air
 - **Layer:** Frontend (UX)
 - **Steps to Reproduce:**
 1. Navigate to `/air`
 2. View the AI Brief section with two similar-looking buttons
 - **Expected:** Clear visual distinction between “fill only” and “fill + auto-run”
 - **Actual:** Both buttons have similar styling (outlined vs filled), making it unclear which is the “safer” choice for users who want to review before submitting.
-

BUG-022: Mobile — AI Brief Panel Text Area Partially Obscured by Microphone Button

- **Severity:**  Medium
 - **Viewport:** Mobile (375×812)
 - **Pipeline:** Road, Air (AI Brief panel)
 - **Steps to Reproduce:**
 1. Load `/road` or `/air` on mobile
 2. Tap into the AI Brief text area
 3. Start typing a long shipment description
 - **Expected:** Full text area with no overlap
 - **Actual:** The microphone icon () overlaps the right side of the text area, potentially covering the last few characters of typed text.
-

LOW SEVERITY BUGS

BUG-023: Footer Copyright Year Static

- **Severity:**  Low
 - **Pipeline:** Cross-cutting
 - **Steps to Reproduce:** Scroll to footer on any page
 - **Expected:** Dynamic year or explicit “2026” matching the current year
 - **Actual:** Shows “© 2026 LogiFlow — Google Solution Challenge 2026” — correct for now, but hardcoded.
-

BUG-024: Road Pipeline — “Share Route” Copies Google Maps URL, Not LogiFlow URL

- **Severity:**  Low
 - **Pipeline:** Road
 - **Steps to Reproduce:**
 1. Get road results, click “Share Route”
 2. Paste clipboard contents
 - **Expected:** LogiFlow shareable link that preserves route parameters
 - **Actual:** Copies a Google Maps URL. Other users won’t see the LogiFlow-optimized view.
-

BUG-025: Privacy Policy Page — Back Button Says “Back to home”

- **Severity:**  Low
 - **Pipeline:** Cross-cutting (Legal)
 - **Steps to Reproduce:**
 1. Navigate from login page → Terms of Service → Privacy Policy
 2. Click “Back to home”
 - **Expected:** “Back to home” or “Back to previous page”
 - **Actual:** “← Back to home” always navigates to `/`, even if the user came from `/login`. Should either use browser back or be labeled more accurately.
-

BUG-026: Road Pipeline — ML Model Metrics Could Benefit from Contextual Help

- **Severity:**  Low
 - **Pipeline:** Road
 - **Steps to Reproduce:** View the ML metrics section showing Delay Detection Accuracy (73%), High-Risk Shipment Detection (68%), Cross-Validated Reliability (78.3%)
 - **Expected:** Tooltips or info icons explaining what these metrics mean for non-technical users
 - **Actual:** Raw ML metrics displayed without context. A logistics manager won’t know if 73% accuracy is “good enough.”
-

BUG-027: Air Pipeline — Route Card Confidence Badges All Show Same Value

- **Severity:**  Low (related to BUG-007)
 - **Pipeline:** Air
 - **Steps to Reproduce:** Run any Air search — all route cards show “94%” confidence
 - **Impact:** While the confidence computation in the frontend is well-implemented for Road (see [RouteResults.tsx L122-168](#)), the Air pipeline appears to use a backend-computed confidence that doesn’t differentiate routes.
-

BUG-028: Legal Pages — No Table of Contents for Long Document

- **Severity:**  Low
 - **Pipeline:** Cross-cutting (Legal)
 - **Steps to Reproduce:** Navigate to `/privacy` or `/terms`
 - **Expected:** Table of contents or section navigation for long legal documents
 - **Actual:** Long scrolling document with no quick navigation
-

BUG-029: Road Pipeline — Route Badge Labels Could Overlap on Mobile

- **Severity:**  Low
 - **Viewport:** Mobile
 - **Steps to Reproduce:** View road results on mobile with multiple badge labels (cheapest, fastest, recommended)
 - **Expected:** Badges should wrap gracefully
 - **Actual:** Badges may overlap when multiple labels apply to the same route on narrow screens
-

BUG-030: Air Pipeline — “Save Report” Button Visible but Disabled Styling is Unclear

- **Severity:**  Low
 - **Pipeline:** Air
 - **Steps to Reproduce:**
 1. Run air search, get results
 2. Click “Save Report” while not logged in
 - **Expected:** Clear disabled state or “Sign in to save” tooltip
 - **Actual:** Button text includes “(sign in required)” but the button itself appears clickable. Compare with Road pipeline where the Save Report has clearer styling.
-

ARCHITECTURAL OBSERVATIONS

ARCH-001: Backend Validation Asymmetry

The Road pipeline has robust validation: - Same-city check  - Frontend weight validation  - Backend Pydantic field validators 

The Air pipeline has **minimal** validation: - No same-city check  - No frontend weight/city validation  - No Pydantic field validators 

Recommendation: Standardize validation across all pipelines. Create a shared `BasePipelinePayload` Pydantic model with common validators.

ARCH-002: State Persistence Between Searches

When switching from one search to another (e.g., valid Air search → gibberish search), stale results from the previous search persist in the UI. The `handleOptimize` function in the store [useLogiFlowStore.ts L384](#) clears results upon **success** but for `no_routes` cases, old results may still be visible in the DOM.

ARCH-003: Confidence Score Design Difference

- **Road:** Frontend-computed using a sophisticated 4-factor weighted formula ([RouteResults.ts x L122-168](#))
- **Air:** Backend-computed, appears to be static (always 94%)

This inconsistency means the confidence metric has different meanings across pipelines.

✓ THINGS WORKING WELL

Feature	Status	Notes
Road form validation (same-city)	✓ Pass	Clean error message with red border
Road multi-stop routing	✓ Pass	Up to 10 stops, reorder, auto-optimize
Road simulation mode	✓ Pass	5 presets, 3 sliders, integrates with pipeline
Road cost breakdown	✓ Pass	7 line items with emoji icons
Road "Start Driving" (Google Maps)	✓ Pass	Opens correct corridor
Road map with route polylines	✓ Pass	Interactive, zoomable, route labels
Air form layout	✓ Pass	Clean, intuitive grid layout
Air "No routes" messaging	✓ Pass	Clear message with rail/road alternative links
Login page design	✓ Pass	Clean, professional, Google OAuth prominent
Login → Terms/Privacy links	✓ Pass	Working navigation
Protected route redirects	✓ Pass	Dashboard/Reports redirect to login
Mobile hamburger menu	✓ Pass	Opens with all nav links accessible
Mobile form stacking	✓ Pass	Forms stack vertically, readable
Homepage hero section	✓ Pass	Beautiful glassmorphic design
AI Brief panel (mobile)	✓ Pass	Usable, buttons properly stacked
Legal pages content	✓ Pass	Comprehensive, well-written
NavBar responsiveness	✓ Pass	Transitions from full links → hamburger



RECOMMENDED FIX PRIORITY

Immediate (Pre-Demo)

1. **BUG-001** — Add same-city validation to Air backend
2. **BUG-002** — Add weight validation to Air frontend + backend
3. **BUG-003** — Clear stale results on new search, deduplicate autocomplete
4. **BUG-005** — Add `scrollIntoView()` after results load

Before Submission

5. **BUG-006/007** — Investigate Air pipeline route generation (all identical routes is suspicious)
6. **BUG-008** — Either implement notification dropdown or remove the bell icon
7. **BUG-010** — Deduplicate city autocomplete results
8. **BUG-017** — Ensure cost breakdown total matches route.cost

Nice-to-Have

9. **BUG-015** — Add tooltip to Window slider
10. **BUG-020** — Move Simulation Mode behind advanced toggle
11. **BUG-012** — Increase touch targets to 44px minimum
12. All  Low severity items

Report generated by exhaustive browser-based testing across desktop (1280×800) and mobile (375×812) viewports, combined with source code review of frontend components and backend route handlers.